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Ansible for z/OS

Working with VSAM Data Sets

TEKsystems Global Services

WORKFORCE DEVELOPMENT

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VSAM DATA SETS

ESDS – Entry-Sequenced Data Set

Records are stored in the order they are loaded and retrieved sequentially.

RRDS – Relative Record Data Set

Records are stored in the order they are loaded and retrieved by Relative Record Number (RRN).

KSDS – Key-Sequenced Data Set

Records are stored in order by logical key and can be browsed or accessed directly.

LDS – Linear Data Set

Processed as a byte stream. Used in z/OS system services, not in applications.

VSAM stands for Virtual Storage Access Method.

For many operational purposes, VSAM supplanted the original, more primitive access methods.

In an infrastructure engineer role, you won't need to understand the internals of VSAM. You will probably have to provision environments that include VSAM data sets.

For backward compatibility, z/OS still supports the BSAM and BDAM access methods for sequential and random record access respectively, but nearly all applications use QSAM or VSAM ESDS for sequential processing, and VSAM KSDS or RRDS for random access.

VSAM LDS data sets are used for the system spooler, DB2, and other z/OS services. Applications don't use LDS, and high-level application languages like COBOL have no support for it. LDS is the only native byte-stream data set type on z/OS.

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VSAM CLUSTERS

VSAM data sets are called "clusters" because they consist of more than one component, each with a separate catalog entry.

ESDS, RRDS, and LDS

- Catalog entry for the cluster
- Catalog entry for the data component

KSDS

- Catalog entry for the cluster
- Catalog entry for the data component
- Catalog entry for the index component

AIX (alternate index)

- Catalog entry for the cluster
- Catalog entry for the data component
- Catalog entry for the index component
- Catalog entry for the path component

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DEFINE VSAM ESDS CLUSTER

```
//IBMUSERE JOB ,'DEFINE ESDS',
//         CLASS=A,MSGCLASS=X,
//         MSGLEVEL=(1,1),NOTIFY=&SYSUID
//*************************************************
//* Define a VSAM ESDS
//*************************************************
//DEFESDS  EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSOUT   DD SYSOUT=*
//SYSIN    DD *
 DELETE IBMUSER.TEST.ESDS
 IF LASTCC < 9 THEN -
   DEFINE CLUSTER( -
     NAME(IBMUSER.TEST.ESDS) -
     RECORDSIZE(80 80) -
     CYLINDERS(2 1) -
     CISZ(4096) -
     NONINDEXED -
   ) -
   DATA( -
     NAME(IBMUSER.TEST.ESDS.DATA) -
   )
/*
```

We use the IDCAMS utility to define VSAM clusters. There are other ways, but this is the most common.

You can see the structure of the job is the same as for defining a GDG. The particular IDCAMS commands are different.

DEFINE CLUSTER pertains to VSAM data sets.

The DSN will be the name of the cluster entry in the catalog.

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     RECORDSIZE(80 80) -
     CYLINDERS(2 1) -
     CISZ(4096) -
     NONINDEXED -
   ) -
   DATA( -
     NAME(IBMUSER.TEST.ESDS.DATA) -
   )
/*
```

VSAM data sets are always treated as having variable-length logical records. For RECORDSIZE, we specify the *average* and *maximum* lengths.

NONINDEXED is the keyword that identifies this cluster as an ESDS as opposed to any of the other VSAM data set types.

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   DEFINE CLUSTER( -
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     CISZ(4096) -
     NONINDEXED -
   ) -
   DATA( -
     NAME(IBMUSER.TEST.ESDS.DATA) -
   )
/*
```

The name of the data component will be the same as the name of the cluster with ".DATA" appended.

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ANSIBLE AND VSAM

Module zos_data_set is a wrapper around basic z/OS data set allocation services.

Several z/OS utilities accept a command file, usually defined as SYSIN, to control sophisticated program features.

Some batch job steps may perform significant processing this way, running IDCAMS, DFSORT, EIBGENER, EIBCOPY, and other z/OS utilities.

zos_data_set has no built-in support for these command files. It can define a VSAM data set, but only using bare-bones specifications and accepting many default values that may not be what you want.

To control the more sophisticated z/OS utilities, you need to use module zos_mvs_raw. The module gives you great flexibility, but you must know how to do "all the things" on the z/OS system in order to use it effectively. Ansible cannot abstract the details away, when you need this level of control.

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LAB 26: DEFINE A VSAM KSDS CLUSTER

Follow the instructions for Lab 26 in labs/labs.md in the course repo.

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WHAT YOU KNOW AND WHAT YOU CAN DO

- You can define VSAM assets using JCL and Ansible

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WHAT'S NEXT

- ICEMAN and DFSORT